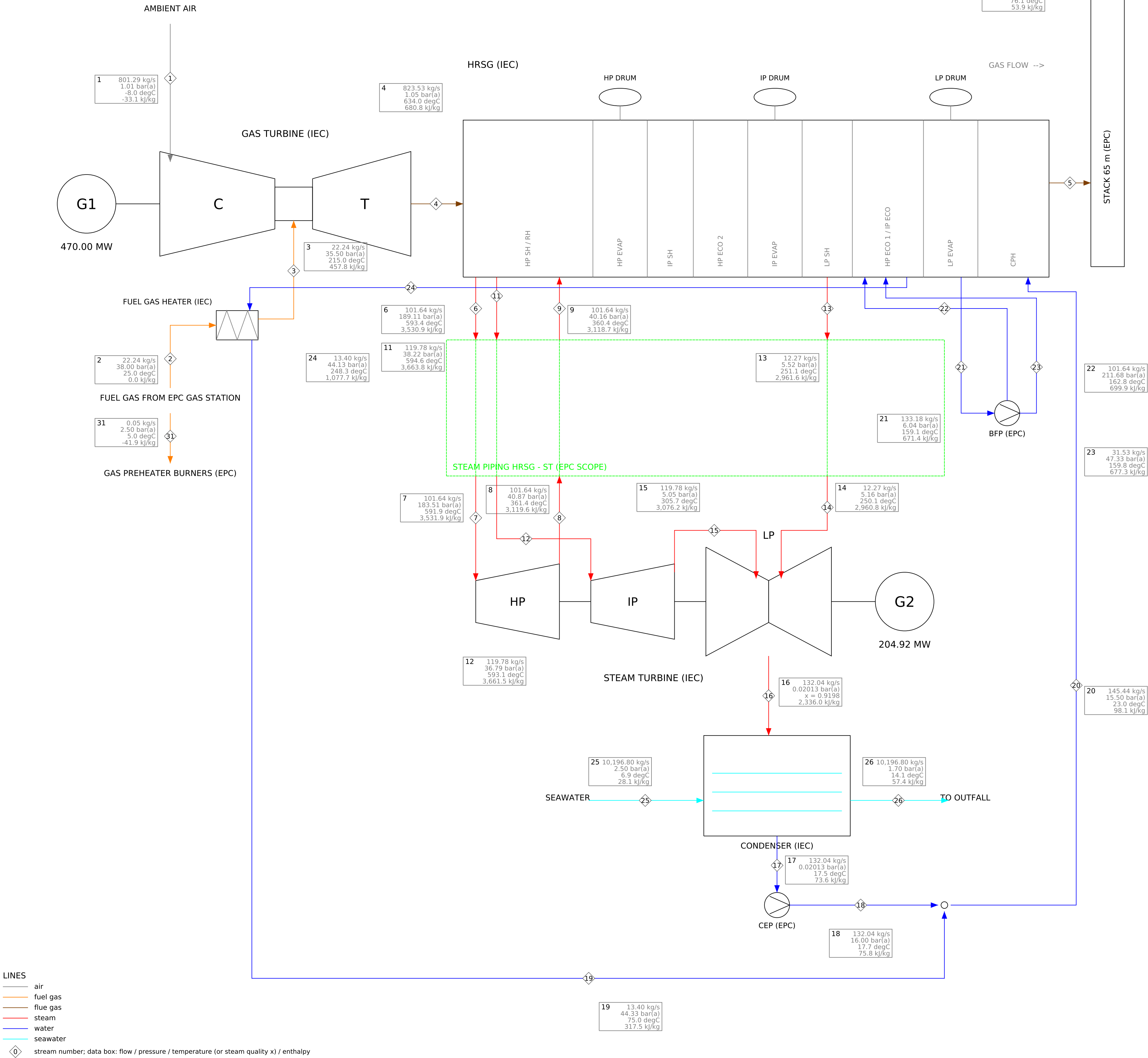




STREAM TABLE - MIN-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	801.29	1.01	-8.0	-33.1	-26.5
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	22.24	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	22.24	35.50	215.0	457.8	10.2
4	GT exhaust gas to HRSG inlet	823.53	1.05	634.0	680.8	560.7
5	Flue gas at HRSG outlet to stack	823.53	1.01	76.1	53.9	44.4
6	HP steam at HRSG outlet (HP SH outlet)	101.64	189.11	593.4	3,530.9	358.9
7	HP steam at ST main stop valve inlet	101.64	183.51	591.9	3,531.9	359.0
8	Cold reheat steam at HP turbine exhaust	101.64	40.87	361.4	3,119.6	317.1
9	Cold reheat steam at HRSG reheater inlet	101.64	40.16	360.4	3,118.7	317.0
10	IP superheated steam (joins cold reheat inside the HRSG)	18.13	41.02	329.4	3,038.2	55.1
11	Hot reheat steam at HRSG outlet	119.78	38.22	594.6	3,663.8	438.8
12	Hot reheat steam at ST IP stop/control valve inlet	119.78	36.79	593.1	3,661.5	438.6
13	LP steam at HRSG outlet	12.27	5.52	251.1	2,961.6	36.3
14	LP steam at ST LP admission	12.27	5.16	250.1	2,960.8	36.3
15	IP turbine exhaust (crossover, internal)	119.78	5.05	305.7	3,076.2	368.5
16	LP turbine exhaust to condenser (UEEP)	132.04	0.0201	x 0.920	2,336.0	308.5
17	Condensate at hotwell outlet / CEP suction	132.04	0.0201	17.5	73.6	9.7
18	Condensate at CEP discharge	132.04	16.00	17.7	75.8	10.0
19	Fuel gas heater water return to condensate	13.40	44.33	75.0	317.5	4.3
20	Condensate to HRSG condensate preheater inlet	145.44	15.50	23.0	98.1	14.3
21	Feedwater from LP drum to boiler feed pumps	133.18	6.04	159.1	671.4	89.4
22	HP feedwater at BFP discharge to HP economiser	101.64	211.68	162.8	699.9	71.1
23	IP feedwater at BFP interstage to IP economiser	31.53	47.33	159.8	677.3	21.4
24	IP economiser water to performance fuel gas heater	13.40	44.13	248.3	1,077.7	14.4
25	Circulating seawater to condenser inlet	10,196.80	2.50	6.9	28.1	286.3
26	Circulating seawater from condenser outlet	10,196.80	1.70	14.1	57.4	585.1
31	Fuel gas to the station preheater burners (water-bath heaters)	0.05	2.50	5.0	-41.9	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	470.00	MW
ST output (generator terminals)	204.92	MW
Gross output	674.92	MW
Auxiliary loads	-9.97	MW
Transformer losses (GSU, UAT, MV/LV)	-2.15	MW
NET OUTPUT (HV terminals)	662.80	MW
Heat input (LHV)	1,058	MW
Gross heat rate (LHV)	5,633	kJ/kWh
NET HEAT RATE (LHV)	5,748	kJ/kWh
Net efficiency (LHV)	62.63	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	109,544	Sm3/h
Circulating water flow	36,047	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	107.96	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.032 MW (tolerance 0.59 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.007 MW (tolerance 1.04 MW)
OK	energy balance HRSG: residual +2.088 MW vs stated casing loss 2.065 MW (tolerance 0.98 MW)
OK	energy balance PIPE: residual +0.274 MW (tolerance 1.15 MW)
OK	energy balance ST: residual -0.003 MW (tolerance 0.83 MW)
WARN	CW temperature rise 7.20 K vs DP-CASE-0015 <= 7.0 K (Condenser cooling water temperature rise)
OK	gas flow 109,544 Sm3/h = 91.3 % of DP-OWNR-0033 (120,000 Sm3/h)
INFO	net export 662.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - MIN-NG-100
CASE	Minimum ambient for full capability (-8 degC), natural gas, base load
DOCUMENT	ALP-HMB-MIN-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	6b60fce553593bf5 2026-09-29
NOTE	all values calculated from the database; do not edit